Research data: the O’Neil chain

Trunk section: §6.5 Research data.

Talks to: Sources (§6.1: SEC EDGAR, Form 13F, Polygon, Yahoo, openFDA, USASpending); the Card (§6.3), which shows the CANSLIM reading; the Market (§6.4), which shows M and L; Quant (§6.8), whose backtest gate reads the same facts as of a day.

Built on: CANSLIM, the deepest document of the old desk: the traps, the exact rules and their provenance, the card and panel specification (locked), the build phases and what each found; qp OPERATIONS (the nightly chain); CARD-FIELDS #28–#88 (what reaches the card). The code: `quant-platform/chart/oneil.py`, `relstrength.py`, `groups.py`, `sic.py`, `edgar.py`, `f13.py`, `base.py`, `ratings.py`, `canslim.py`, and `deploy/run_daily.py`.

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1. What this domain is

William O’Neil’s CAN SLIM is a method for finding leading growth stocks: **C**urrent quarterly earnings, **A**nnual earnings, **N**ew (product, management, industry conditions, price high out of a base), **S**upply and demand, **L**eader or laggard, **I**nstitutional sponsorship, **M**arket direction.

The desk reconstructs it from **free, official data**, names every reconstruction as one (`rs`, never “IBD RS”), and uses it as a **label, never a filter**: no tool’s candidates change because of it. The results are card fields and Market-tab sections; this domain computes them.

Why this domain is separate. Its facts are slow (filings, quarterly 13F, whole-market rankings), heavy (thousands of companies, rate-limited sources) and change at most once a day. They must never be computed inside a page request or a scan, and they must not depend on the chart server being up.

2. The traps the method must avoid

Most CAN SLIM concepts have a plain-English name that means something else in common use. Implementing the common reading produces numbers that look right and measure a different thing. From CANSLIM §1:

Concept	What it is	What it is mistaken for
RS Rating	a percentile 1–99 of 12-month performance against **every other stock**	RSI, which compares a stock to its own range
C	current quarter EPS vs **the same quarter a year ago**	sequential quarters (reintroduces seasonality)
A	annual EPS growth over 3–5 years, plus ROE: durability	a longer C
Distribution day	one session: index down ≥ 0.2 % on **higher volume**	any down day
Distribution count	days **live** in a rolling 25-session window, with expiry	a running total
A/D Rating	a letter A–E from 13 weeks of price and volume	Chaikin's A/D index, an unbounded cumulative line
Follow-through day	a confirmation on **day 4+** of a rally attempt, big gain on higher volume	any strong up day
N: new high	price emerging from a **base** into new ground, on volume	any 52-week high
Group rank	rank of the stock's **industry group**	the sector (far too coarse)
M	a **rule-based state machine** over two indexes	a sentiment read or an MA cross
RS line	the plotted ratio stock ÷ index; "new high **before** price" is the tell	the RS Rating
U/D volume ratio	up-day volume ÷ down-day volume over 50 days	the A/D rating
Base stage	which base counted **from the market bottom**	any consolidation
Group rank vs Group RS	**one fact**: rank 63 of 197 *is* the 68th percentile *is* B+	three measurements that could disagree
RS within the group	"RS 1 of 13": the stock ranked against its own group only	the RS Rating
N: the new thing	**three** things: product, management, **better industry conditions**	two things, and a news feed

3. The legacy method, letter by letter

Every rule below carries over. Thresholds are **O'Neil's own numbers**, named constants, printed beside any signal that fires (CANSLIM §5).

3.1 M: the market model

A state machine over the **S&P 500 (^GSPC)** and the **Nasdaq Composite (^IXIC)**, daily bars. Three published states (confirmed uptrend, uptrend under pressure, market in correction) and one internal (rally attempt).

Rule	Value	Provenance
Distribution day	close ≤ prior close × (1 − 0.002) **and** volume > prior volume **and** the market is in an uptrend	IBD's −0.2 % floor on O'Neil's plain "closes lower on higher volume" (set the floor to 0 for the plain form)
Stalling day	volume above the 25-day average or the prior day, gain under ~0.2 %, close in the lower half of the range: counts as distribution, flagged apart	IBD
Live window	25 sessions	IBD; the tight end of O'Neil's "5–6 weeks"
Removal	after 25 sessions, **or** when the index **trades** (intraday high) 5 % above that day's close	IBD
Under pressure	5 live distribution days	IBD
Correction	6+ live, or a decisive break of the rally low	IBD
The bottom	the **lowest close** of the correction (not an intraday low)	the 2010 workshop
Rally day 1	the session **after** that lowest close, up or down; a new lower close moves the bottom and restarts the count	the workshop
Follow-through	day **4 or later**, gain **≥ 1.7 %**, volume **> the prior session** (that is the whole test)	O'Neil's last figure (1 %, then 1.5 %, then 1.7 %)
Late follow-through	days 8–11 flagged late, beyond 11 very late: **flagged, never refused**	refusing creates a state the market can never leave
Status	counted **per index**, published as the **worse** of the two	"one index is enough"

Lesson recorded from the first live run. An extra test (volume above the 50-day average) blocked the Nasdaq's August 2026 follow-through; the rally ran to day 23 unconfirmed and the

model said “correction” through a confirmed rally. A rule stricter than the published one fails to the wrong answer, in the direction of telling you not to buy. The 50-day comparison is reported beside a follow-through, never decides it.

Caveat, always shown: about one follow-through in four fails. It is necessary for a bottom, not sufficient.

3.2 RS: the relative strength rating

```
RS = 0.4·(P0/P63) + 0.2·(P0/P126) + 0.2·(P0/P189) + 0.2·(P0/P252)
```

63/126/189/252 trading days = 3/6/9/12 months, the latest quarter weighted double; then **percentile-ranked 1–99 against every US stock**. Verified against IBD’s published formula. Two choices that make it match: the universe is the whole market (Polygon grouped daily, every US ticker each session), and prices are **split-adjusted** (a 2-for-1 split would otherwise score a leader at −50 %).

3.3 L: groups and the four levels of strength

Level	The number	How	Answers
Market	the M state + the index RS line	§3.1; `index ÷ S&P 500`	buy at all?
Sector	rank of a coarse sector (planned: 1 of 33)	aggregate performance	which half of the market
Group	rank X of N	**median** member RS, membership by SIC	is this the leading group (O’Neil: ~37 % of a stock’s move; sector ~12 %)
Stock in group	“RS 1 of 13”	the stock’s RS ranked inside its group	leader or laggard

Membership is the SIC code from EDGAR for every filer (TradingView labels only where SIC has none, F-21). **Thin groups** (fewer than 6 ranked members, `QP_GROUP_MIN_MEMBERS`) roll up one rung: 4-digit industry → 3-digit industry group → 2-digit major group; dropped if still thin. Groups are ranked by median RS; **percentile and letter** (A+ ≥ 94 … E) are derived from the rank, never stored as separate facts. **Rotation**: rank now against rank 3 months ago. **The divisor is stored with every rank** (IBD once went from 197 groups to 145); the card prints `63 of 197`, never a bare rank. **The honest limit**: our group rank is ours, not IBD’s, because membership differs; the rank within the group and the member count are exact.

3.4 C and A: earnings, from EDGAR

From `data.sec.gov/api/xbrl/companyfacts/CIK{…}.json` (official, free, a descriptive User-Agent; CIK ↔ ticker from `company_tickers.json`).

C: eight quarters of EPS, year-ago EPS, %Chg, sales, year-ago sales, %Chg, after-tax margin; **accelerating** (each of the last 2–3 quarters' %Chg larger than the one before); **beat +25 %** X of 8. **A**: fiscal years of EPS and %Chg (with the year's price high and low), the year **rebuilt from four quarters** to check against the filing, **3-year growth**, **stability** (the wobble around trend, 0–99, low is good), **ROE** against O'Neil's **17 %** floor.

The five EDGAR traps, each of which produces a normal-looking wrong number (CANSLIM §17.1):

#	Trap	Rule
1	C is year over year, never sequential	compare to the quarter one year back (within 45 days)
2	Q4 is usually not filed as a quarter	Q4 = FY − Q1 − Q2 − Q3, **only when all three are present**; when EPS and net income disagree in sign, Q4 is left out
3	the same period is filed more than once	the most recently **filed** value wins
4	a % from a negative base is not a number	`n/a (loss a year ago)`; capped at ±999 %
5	revenue lives under four tags	read all four, take one per period

Filings land weeks after quarter end, so **every block carries its own as-of date**.

3.5 N: the base, and what is new

The weekly base (weekly is not a display choice: a base is 7–65 weeks, and daily bars drown every shape test in noise): weeks, depth, waves down (the decline typically comes in three), **accumulation weeks** (heavy volume, no price progress), **handle** (a controlled 2–8-week drift in the **upper half** of the base: a handle that gives back half the cup is the cup failing), lip and low with their weeks, **pivot**, % away, X of 6 checks. Needs 7 weekly bars.

The RS line tell: the RS line at a new high **while price is still in its base** fires; both at new highs does not (that is arithmetic); the line off its high does not. **Divergence**: the index made a new low in the window, the stock did not, and the stock is above where it sat on the index's worst close; "the index has not made a new low" is shown as a different answer from "it followed the index down".

N, what is new (designed, phase 7, not built): see §4.

3.6 S: supply

Shares outstanding (EDGAR `dei:EntityCommonStockSharesOutstanding`, its basis when not the cover figure, as-of); **share count change over about a year**, always from the same tag (buying back / diluting / flat, or the reason there is no rate); float (TradingView) and float as % of shares; short interest and days to cover (from the card's short-interest facts); **U/D volume**

ratio over 50 sessions; **A/D**: a bounded 13-week score (−1 … +1, normalised by the volume in the window, comparable between stocks), banded to a letter that is **ours** (IBD does not publish its boundaries), with the raw number beside it.

3.7 I: institutional sponsorship, from Form 13F

The SEC's quarterly 13F datasets. Counted as distinct **managers** by CIK, unioned across every CUSIP of the issuer (stock, converts, options are one company) and across a 13F-HR and its 13F-HR/A (one manager). Per quarter: the holder count, **of N** managers who filed at all, and the **share of filers**. **Direction is read from the share**, with a 3 % relative dead band: `rising` (up over the year and still up), `peaked` (up over the year, newest quarter down), `flat`, `falling`, `turning up`; only `rising` and `turning up` are coloured bullish; the raw holder change is printed beside, uncoloured. Only managers over $100M file, 45 days after quarter end: the lag is stated on screen.

3.8 On each card: the market reflected

From CANSLIM §14, four per-card readings, each a different number on every card:

1. **Behaviour on the live distribution days**: on how many the stock's close-to-close return beat the index's, and the average difference, with the dates behind a tap; says so when there are none.
2. **Sessions since the follow-through**, banded early (< 25), established (25–150), late (> 150).
3. **Group rotation**: rank now vs 3 months ago, ▲ into / ▼ out of.
4. **The exposure implication**, in O'Neil's words.

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4. What was built, and what was not

From the build manifest (CANSLIM §12, §16):

Phase	Delivered	State
1	M: the whole market model (M1–M10)	built
2	M on the Market tab and its per-card reflection	built
3	groups: RS, Grp, volume vs 50-day, `grp X of N`, `RS X of N`, the group table, rotation, % off high	built
4	A/D, U/D ratio, RS line tell, divergence	built
5	EDGAR C and A tables, shares outstanding, the weekly base	built
6	13F I and FINRA S	I built; float estimate (S2) and funds / banks / management % (I3) not built
7	**the N pipeline**	**not built**
—	Composite (R1), EPS rating (R2), SMR (R5)	deferred: a composite without EPS and SMR would be a partial one wearing a complete name
—	group new highs / lows (L4), sector rank (L7), index RS line (M11)	not built

The N pipeline, as designed (CANSLIM §10):

```
HARVEST (24 months) → CLASSIFY (3 buckets) → REACT (price) → PERSIST (RS line) → SCORE → one line
```

Step	Rule
Harvest	EDGAR 8-K **Item 5.02** (new management: structured, the item code is the classifier), **8.01 / 7.01** (product announcements), **1.01** (material agreements); **openFDA** approvals, 510(k), PMA; **USASpending** contract awards; the group's rank rising (new industry conditions, computed). News APIs corroborate, never trigger
Classify	keep new product/service, new management, new contract, better industry conditions. **Drop** analyst notes, index changes, earnings releases (that is C and A), splits, buybacks, dividends, lawsuits, short reports. **Route** secondary offerings to S
React	on the event date + 2 sessions: the move in multiples of the stock's 20-day ATR, volume against its 50-day average. **No move beyond 1 ATR on normal volume: dropped**
Persist	the RS line today against the event day: higher 1.0, flat 0.5, lower 0.2
Score	`N_event = W(class) × R × D(age) × P`; N = the best event, 0–100. Class weights: product 1.00, management 0.90, contract 0.80, industry conditions 0.70, a new high alone 0.40. Age decay: 0–3 months 1.0, 3–6 months 0.9, 6–12 months 0.7, 12–24 months 0.4, older 0. Dedupe by class and date ± 3 sessions
Show	one line (`N 86 · New CEO · 2026-03-12 (5.8 mo) · +11.4 % on 4.2× vol · RS line held`); the event table; **unconfirmed** events (too new to have reacted) listed apart; **discarded counts printed**, as proof that it is a filter, not a feed

5. What failed

ID	Fault	Root cause	Prevented by
—	a ticker first seen while qp is down or restarting has a blank CANSLIM reading until a later scan, and a photo taken then freezes empty `cs*` columns into R1; five tools ask qp for the same tickers	the chain lives inside the chart server; each tool caches its answers for the day	the research worker writes facts; cards read them
—	a new ticker’s C and A arrive only if a page shows its card (a background fetch, one name at a time) or the nightly walk reaches it	lazy fetching tied to the page	the worker’s new-ticker run (§6.1), whatever is on screen
Q-N5	the nightly job is a systemd timer installed by hand with a hard-coded user and path; without it the data goes stale silently	the one scheduler outside pm2	the platform scheduler runs it; a missed or failed run alerts
F-21 (fixed)	the industry map flipped between SIC and TradingView every morning	two writers for one file	one writer: the research worker
—	a missing O’Neil file made the Market tab ask qp to build it inside a page request	computation in the request path	pages only read; “not built yet · last run DATE”
—	the first live data check reported a different environment from the running one (a bare shell without the keys)	the check did not load the service’s environment	health checks run inside the worker’s own environment
D-D3	R2’s O’Neil status read at export time	not frozen	the market state carries the M fact as it was
—	the backtest rebuilds CANSLIM “as of” a day separately	no dated facts	the gate reads facts by `as_of`
—	each piece rebuilt on first request when the job did not run, so the first person in the morning paid for it	lazy builds as a fallback	no lazy builds; the scheduler is the only trigger

6. The new design

6.1 The research worker

A Python job (it reuses qp's verified code: `relstrength.py`, `oneil.py`, `groups.py`, `edgar.py`, `f13.py`, `base.py`, `ratings.py`), always started by the platform scheduler. It runs its steps, writes facts, records its run, and exits. Each step is independent: one failure never stops the rest, and each failure is reported.

Run	When (New York, trading days)	Steps
the night	18:30, once the day's whole-market bars are published (today's timer: 22:30 UTC)	all eleven, below
new tickers	every 5 minutes, 04:00–16:00, while any ticker carded today has no company facts and no attempt today	6–10 for those tickers only
provisional M	every 30 minutes, 09:30–16:00	3, on today's index bars so far, written as `provisional`; the night's run replaces it
by hand	a button or `desk research <step>`: it queues the job, and no page waits	any

The "new tickers" list is a query over today's cards and the facts (a card whose ticker has no company fact and no attempt today), not a queue table, so the worker stays the only writer. EDGAR allows ten requests a second, so a morning's new tickers land within minutes; until then the card says "computing · since 09:41".

#	Step	Writes (fact family)	Clock	Budget
1	RS universe top-up	`bars_universe` (Polygon grouped daily, the newest sessions)	each close	minutes
2	RS ratings	`rs_rating` per ticker (with the universe size)	each close	—
3	market model	`oneil_m` for `market` (state, per-index counts, distribution days, rally, follow-through, rules in use)	each close; every 30 minutes in the session as `provisional`	—
4	SIC map	`sic` per ticker	new listings and changes: nightly	—
5	groups	`group_ranking` for `market`; `oneil_l` per ticker (group, level, rank of N, percentile, letter, RS in group, rotation, roll-up)	each close	—
6	earnings	`oneil_c`, `oneil_a` per ticker	weekly, and on a new 10-Q or 10-K	a time budget per night
7	supply	`oneil_s` per ticker	monthly (shares), each close (U/D, A/D)	—
8	institutions	`oneil_i` per ticker	quarterly, after the 45-day deadline	—
9	base and RS line	`oneil_n_base`, `rs_line` per ticker	each close (weekly bars)	—
10	ratings	`oneil_ratings` (RS, Grp, A/D, volume vs 50-day; then EPS, SMR, Composite)	each close	—
11	N events	`oneil_n_events` per ticker	nightly, 24-month window	a time budget

Which tickers. Whole-market steps (1–5) cover every US stock, so RS, the group and the SIC code exist for any ticker the moment it is carded. Per-company steps (6–11) cover, each night, the union of every ticker any tool has carded in the last 90 days, every R1 ticker and every ticker in an open trade; a ticker carded for the first time is covered the same morning by the new-ticker run.

6.2 Facts, with dates

Each fact family above is written to the fact store with:

- `as_of`: the date it describes (the quarter end, the session, the 13F quarter);
- `known_at`: when the worker learned it;

- `status`: `ok`, `none` (e.g. no filings) or `unavailable` (e.g. EDGAR timed out), with the reason;
- the **rules in use** for anything with a threshold, so a signal can be checked against its definition later.

This makes the backtest's CANSLIM gate exact: it reads the facts known before its decision bar (trunk §5.2).

6.3 Who reads what

Reader	Reads	Never
the card engine	`oneil_*`, `rs_rating`, `rs_line` for the card's ticker	computes any of them
the market engine	the latest `oneil_m` (final or provisional), `group_ranking`	computes any part of them
quant	any of them, as of a day	writes them
pages	through the card and the market state	trigger a build

6.4 What gets built that was not

In order, each behind its own versioned definition:

1. **Index RS line** and **group new highs / lows** (cheap, from data already fetched).
2. **Sector rank** from SIC major groups, kept distinct from the group.
3. **EPS rating and SMR** from EDGAR, then **Composite** once both exist.
4. **Float estimate** (shares outstanding minus insider and 5 %-holder positions from Forms 4 and SC 13D/G), labelled as an estimate; **funds / banks / management %**.
5. **The N pipeline**, as designed in §4.

6.5 Everything explains itself

Each fact family ships with its definition card: what it means, how it is calculated, **what it is not** (from the trap table in §2), source and as-of. The card and the Market tab show it on a tap (CARD.md §5.6).

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7. Contracts

Owns (only writer)

Fact family	Subject
`bars_universe`, `group_ranking`, `oneil_m`	`market`
`rs_rating`, `rs_line`, `sic`, `oneil_l`, `oneil_c`, `oneil_a`, `oneil_s`, `oneil_i`, `oneil_n_base`, `oneil_n_events`, `oneil_ratings`	ticker

Reads

From	What
sources	EDGAR (company facts, tickers, 8-K items, SIC), 13F datasets, Polygon grouped daily, Yahoo index and stock bars, openFDA, USASpending
tools and cards	which tickers to cover
platform	its schedule and environment

Run record

Each run writes a job-run row: start, end, each step's result, counts (tickers covered, facts written, failures with reasons). A step that has not succeeded within its clock (for example, earnings not refreshed in 8 days) turns the health page amber, then red, and sends an alert.

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8. How the rebuild is proven

1. **Same numbers.** For a week the worker runs beside qp's nightly chain; for every ticker and family, its facts are compared with qp's files. Differences are explained or fixed before the card switches to the facts.
2. **qp down test.** With quant stopped, every card's CANSLIM reading and the Market tab's M and L are unchanged.
3. **The traps as tests.** Each row of §2 and each EDGAR trap of §3.4 has a fixture: a negative base, a missing Q4, a restatement, a tag switch, a split, a 13F amendment, a lowest close vs an intraday low, a day-3 bounce, a late follow-through.
4. **M against charts.** One month of distribution days and follow-throughs is checked by hand against index charts.
5. **A missed run.** Skip the worker for a night: the health page goes amber and an alert arrives; cards show "last run DATE".

6. **A new ticker.** A ticker carded for the first time at 10:00 shows its C, A, N and S readings by 10:15, with nothing on screen.